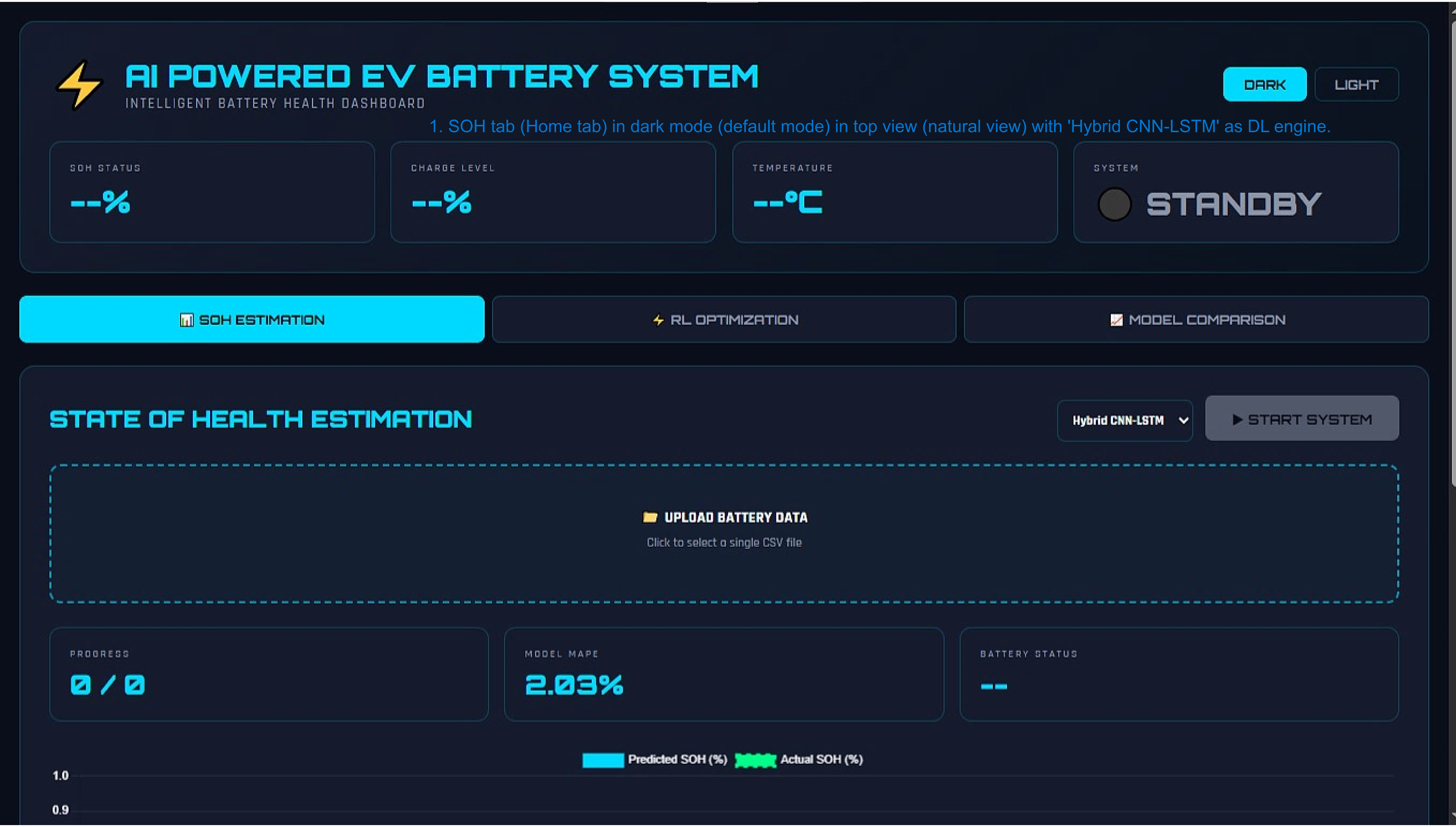

1. SOH tab (Home tab) in dark mode (default mode) in top view (natural view) with 'Hybrid CNN-LSTM' as DL engine.

PROGRESS

0 / 0

MODEL MAPE

2.03%

BATTERY STATUS

--

Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

Team: Md. Shiban Alam, Syed Sarafeena Ali, Purbasha Roy, Dindrila Sain

Mentor: Mrs. Amrita Bhattacharya | **Institution:** STCET - MAKAUT

AI POWERED EV BATTERY SYSTEM
INTELLIGENT BATTERY HEALTH DASHBOARD
DARK
LIGHT
SOH STATUS
--%
CHARGE LEVEL
--%
TEMPERATURE
--°C
SYSTEM
STANDBY
SOH ESTIMATION
RL OPTIMIZATION
MODEL COMPARISON
REINFORCEMENT LEARNING LOGS
Deep Q-Network (DQN)
RECOMMENDED ACTION
--
CUMULATIVE REWARD
0.0
ACTION EFFICIENCY
--%
Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles
Team: Md. Shiban Alam, Syed Sarafeena Ali, Purbasha Roy, Oindrila Sain
Mentor: Mrs. Amrita Bhattacharya | Institution: STCET - MAKAUT
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AI POWERED EV BATTERY SYSTEM
INTELLIGENT BATTERY HEALTH DASHBOARD
DARK
LIGHT
SOH STATUS
--%
CHARGE LEVEL
--%
TEMPERATURE
--°C
SYSTEM
STANDBY
SOH ESTIMATION
RL OPTIMIZATION
MODEL COMPARISON
MODEL PERFORMANCE ANALYSIS
Phase 1: SOH Estimation Models
CNN Model
MAPE 4.09%
R² SCORE -0.280
TRAINING TIME ~ 20 sec
LSTM Model
MAPE 2.62%
R² SCORE 0.476
TRAINING TIME ~ 30 sec
Hybrid CNN-LSTM
MAPE 2.03%
R² SCORE 0.642
TRAINING TIME ~ 35 sec
Phase 2: RL Optimization Algorithms
Q-Learning (Baseline)
Deep Q-Network (DQN)

MODEL PERFORMANCE ANALYSIS

Phase 1: SOH Estimation Models

🧠 CNN Model

MAPE	4.09%
R² SCORE	-0.280
TRAINING TIME	~ 20 sec

📈 LSTM Model

MAPE	2.62%
R² SCORE	0.476
TRAINING TIME	~ 30 sec

🏆 Hybrid CNN-LSTM

MAPE	2.03%
R² SCORE	0.642
TRAINING TIME	~ 35 sec

Phase 2: RL Optimization Algorithms

📊 Q-Learning (Baseline)

IMPROVEMENT	+5.8%
TRAINING TIME	~ 5 min
ARCHITECTURE	Tabular Matrix

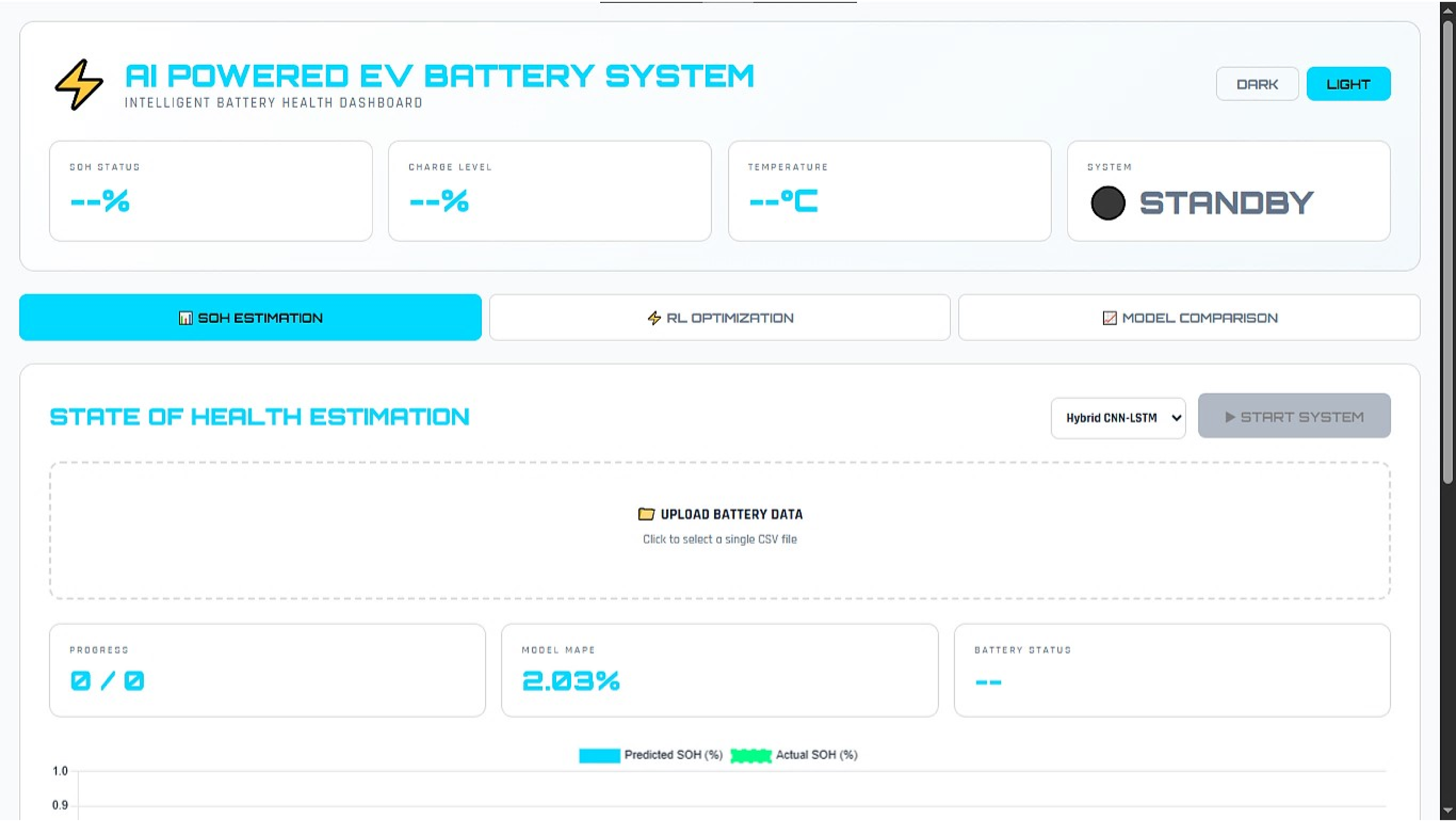
🏆 Deep Q-Network (DQN)

IMPROVEMENT	+12.6%
TRAINING TIME	~ 20 min
ARCHITECTURE	Neural Net (Stable-Baselines3)

Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

Team: Md. Shiban Alam, Syed Sarafeena Ali, Purbasha Roy, Oindrila Sain

Mentor: Mrs. Amrita Bhattacharya | **Institution:** STCET - MAKAUT


AI POWERED EV BATTERY SYSTEM
INTELLIGENT BATTERY HEALTH DASHBOARD
DARK
LIGHT
SOH STATUS
--%
CHARGE LEVEL
--%
TEMPERATURE
--°C
SYSTEM
STANDBY
SOH ESTIMATION
RL OPTIMIZATION
MODEL COMPARISON
STATE OF HEALTH ESTIMATION
Hybrid CNN-LSTM
START SYSTEM
UPLOAD BATTERY DATA
Click to select a single CSV file
PROGRESS
0 / 0
MODEL MAPE
2.03%
BATTERY STATUS
--
Predicted SOH (%)
Actual SOH (%)
1.0
0.9

PROGRESS

0 / 0

MODEL MAPE

2.03%

BATTERY STATUS

--

Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

Team: Md. Shiban Alam, Syed Sarafeena Ali, Purbasha Roy, Dindrila Sain

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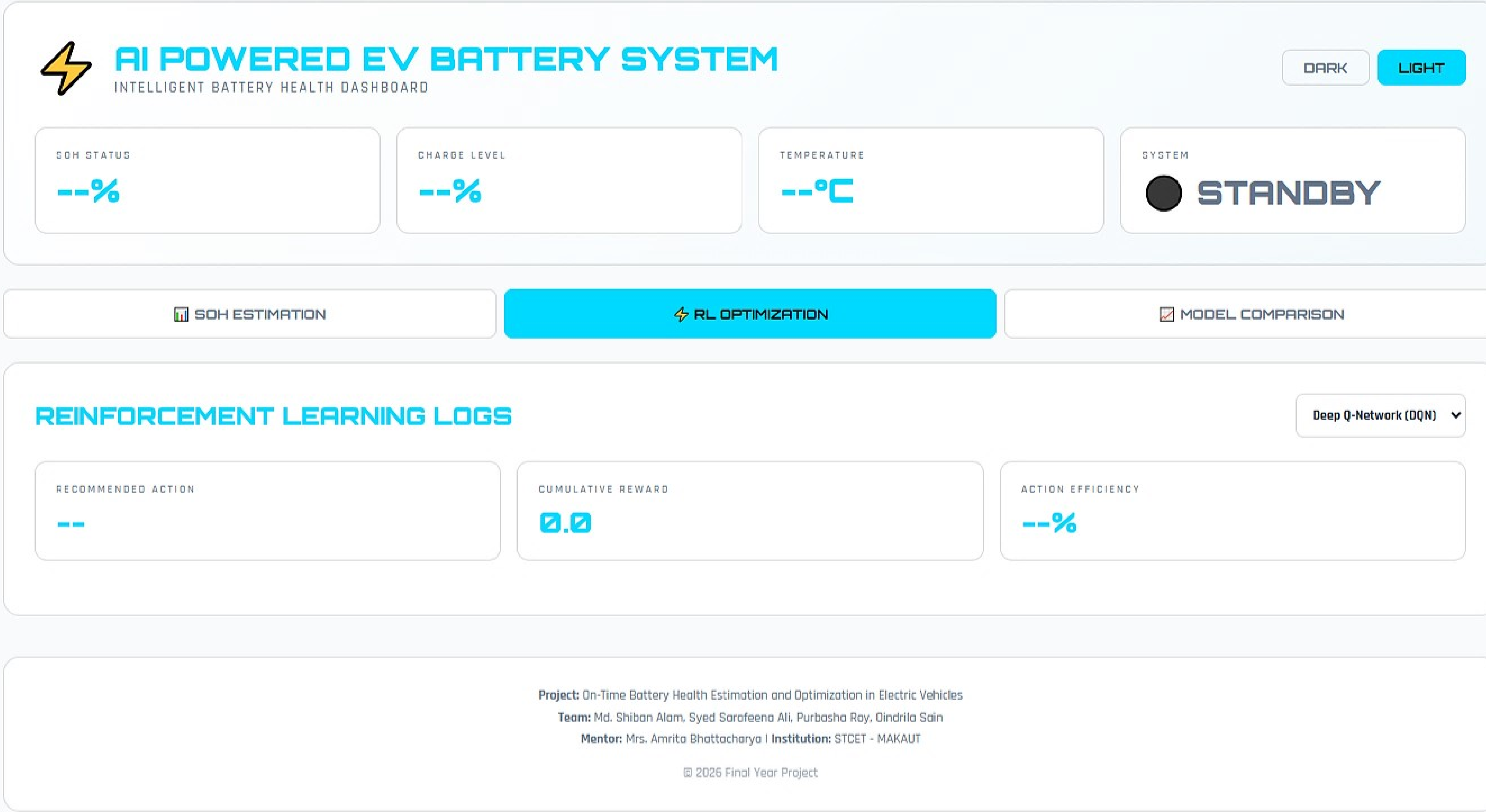

AI POWERED EV BATTERY SYSTEM
INTELLIGENT BATTERY HEALTH DASHBOARD
DARK
LIGHT
SOH STATUS
--%
CHARGE LEVEL
--%
TEMPERATURE
--°C
SYSTEM
STANDBY
SOH ESTIMATION
RL OPTIMIZATION
MODEL COMPARISON
REINFORCEMENT LEARNING LOGS
Deep Q-Network (DQN)
RECOMMENDED ACTION
--
CUMULATIVE REWARD
0.0
ACTION EFFICIENCY
--%
Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles
Team: Md. Shiban Alam, Syed Sarafeena Ali, Purbasha Roy, Oindrila Sain
Mentor: Mrs. Amrita Bhattacharya | Institution: STCET - MAKAUT
2026 Final Year Project

MODEL PERFORMANCE ANALYSIS

Phase 1: SOH Estimation Models

🧠 CNN Model

MAPE	4.09%
R² SCORE	-0.280
TRAINING TIME	~ 20 sec

📈 LSTM Model

MAPE	2.62%
R² SCORE	0.476
TRAINING TIME	~ 30 sec

🏆 Hybrid CNN-LSTM

MAPE	2.03%
R² SCORE	0.642
TRAINING TIME	~ 35 sec

Phase 2: RL Optimization Algorithms

📊 Q-Learning (Baseline)

IMPROVEMENT	+5.8%
TRAINING TIME	~ 5 min
ARCHITECTURE	Tabular Matrix

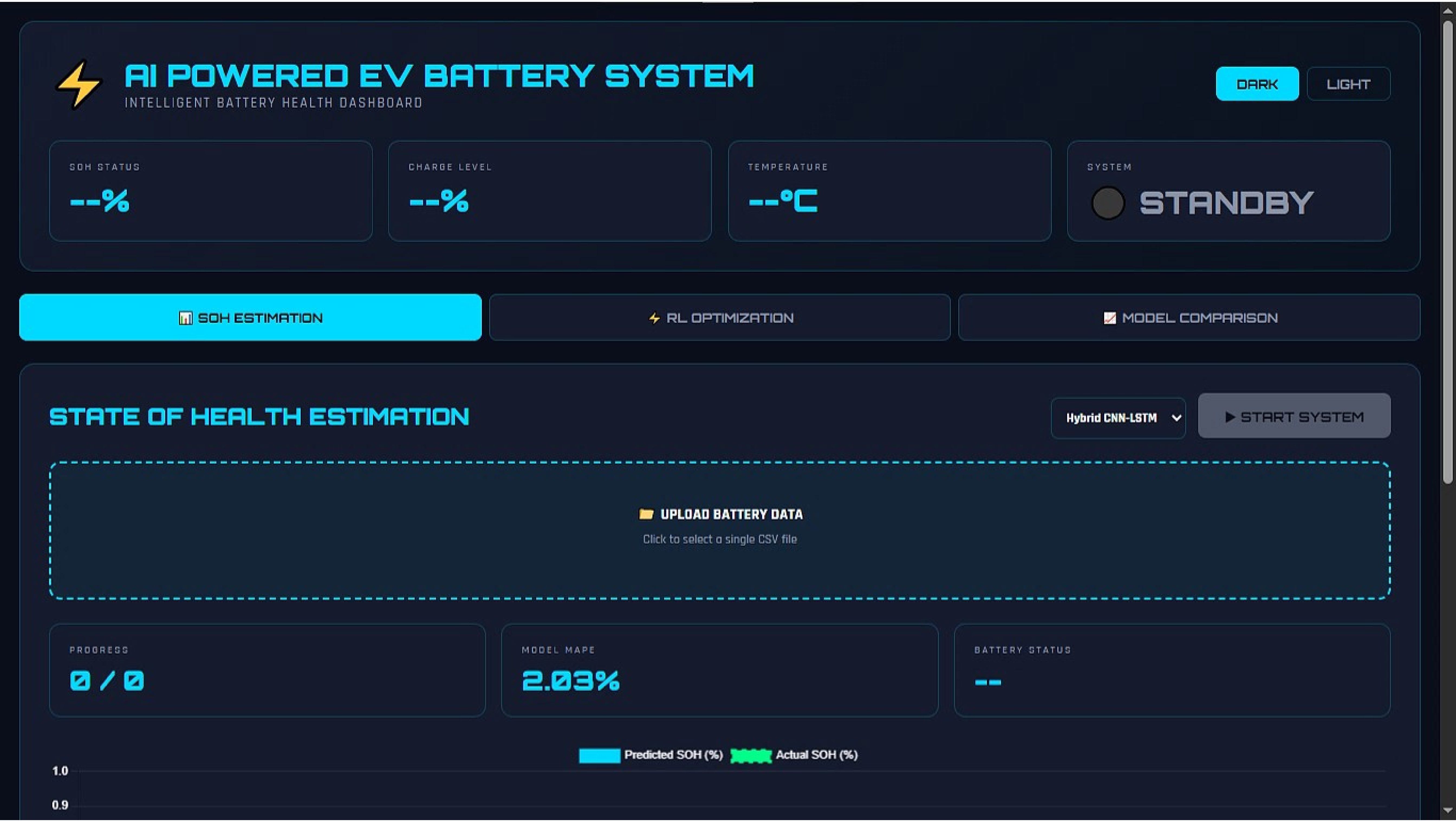
🏆 Deep Q-Network (DQN)

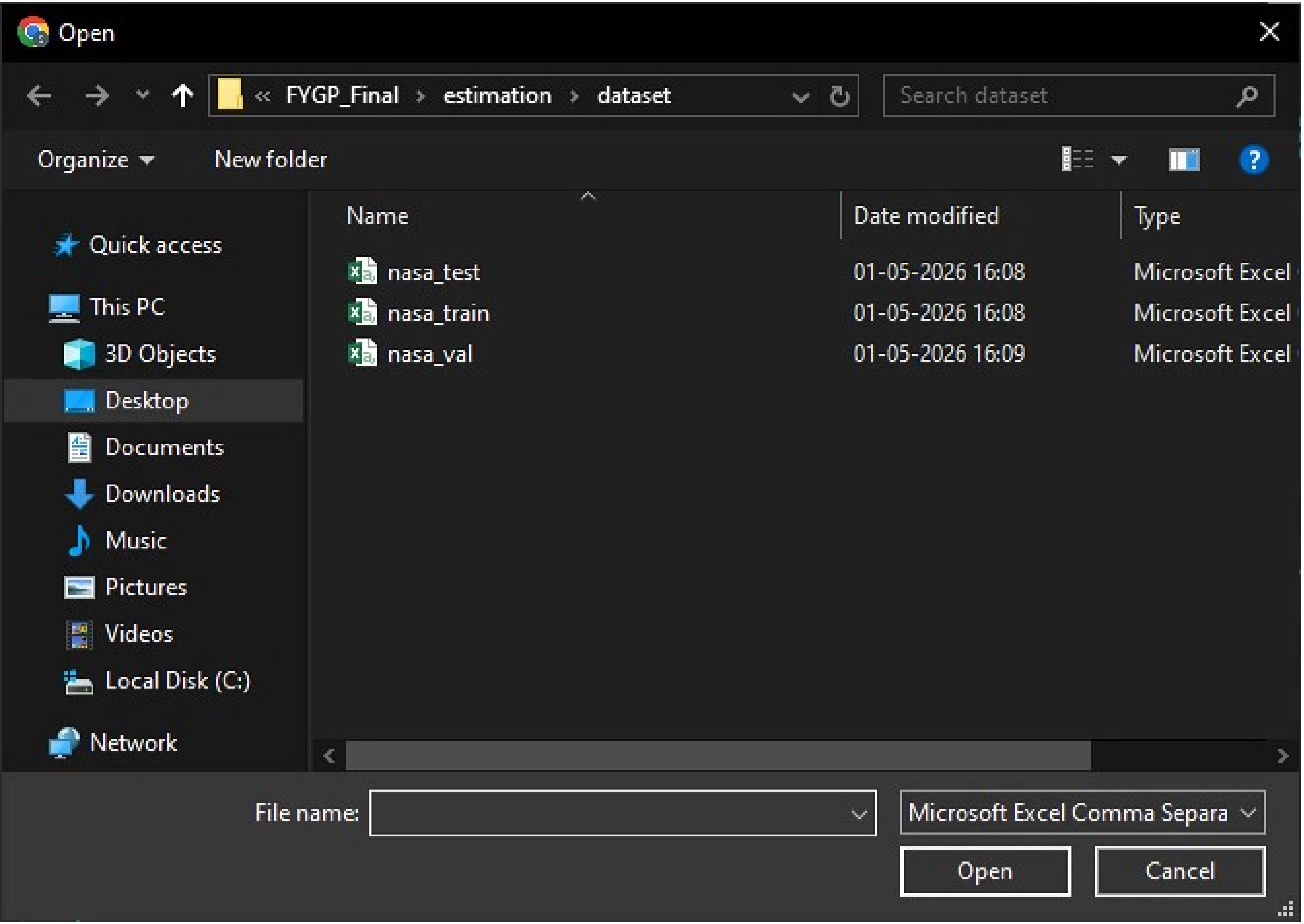
IMPROVEMENT	+12.6%
TRAINING TIME	~ 20 min
ARCHITECTURE	Neural Net (Stable-Baselines3)

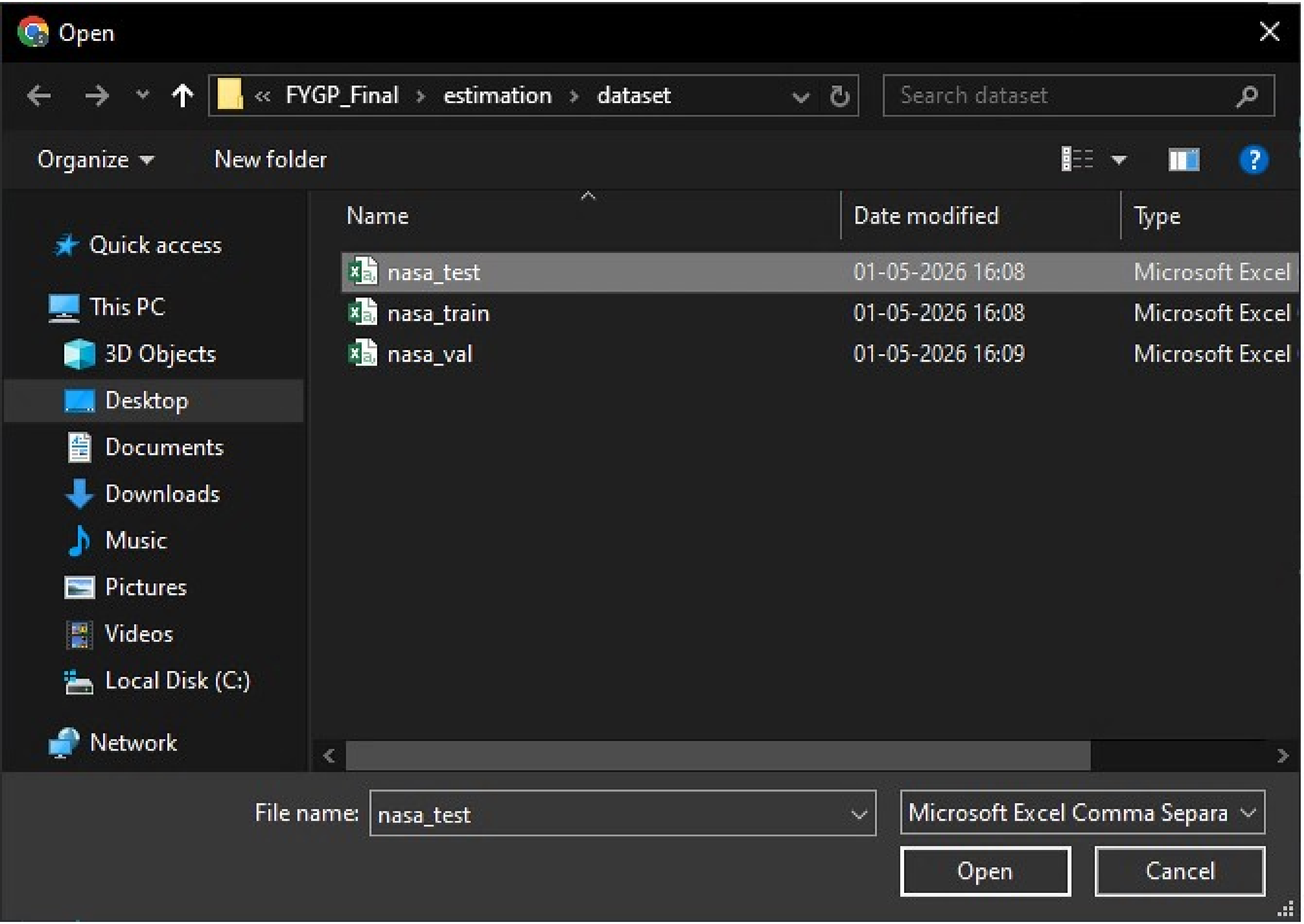
Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

Team: Md. Shiban Alam, Syed Sarafeena Ali, Purbasha Roy, Oindrila Sain

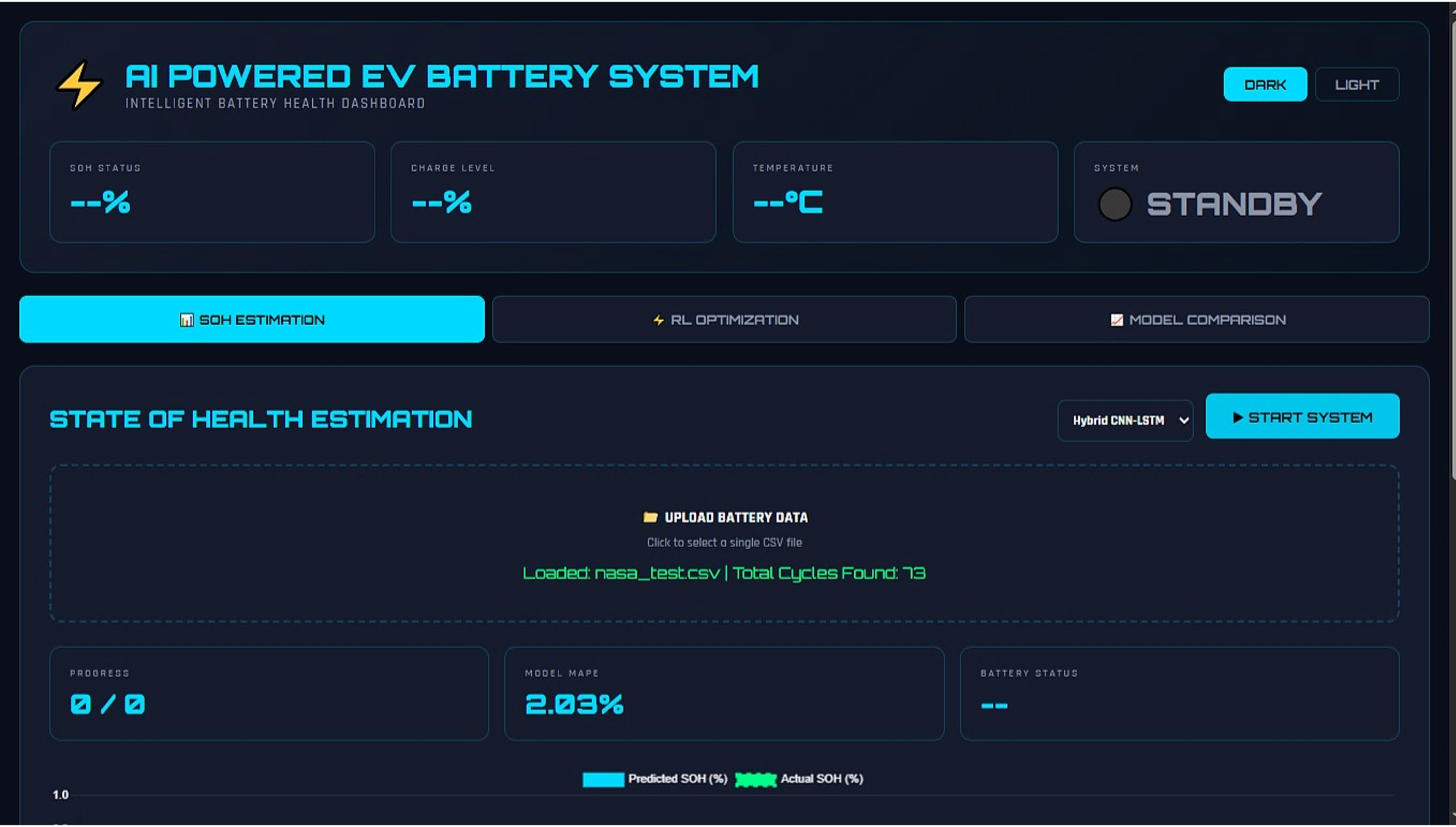
Mentor: Mrs. Amrita Bhattacharya | **Institution:** STCET - MAKAUT


AI POWERED EV BATTERY SYSTEM
INTELLIGENT BATTERY HEALTH DASHBOARD
DARK
LIGHT
SOH STATUS
--%
CHARGE LEVEL
--%
TEMPERATURE
--°C
SYSTEM
STANDBY
SOH ESTIMATION
RL OPTIMIZATION
MODEL COMPARISON
STATE OF HEALTH ESTIMATION
Hybrid CNN-LSTM
START SYSTEM
UPLOAD BATTERY DATA
Click to select a single CSV file
PROGRESS
0 / 0
MODEL MAPE
2.03%
BATTERY STATUS
--
Predicted SOH (%)
Actual SOH (%)
1.0
0.9

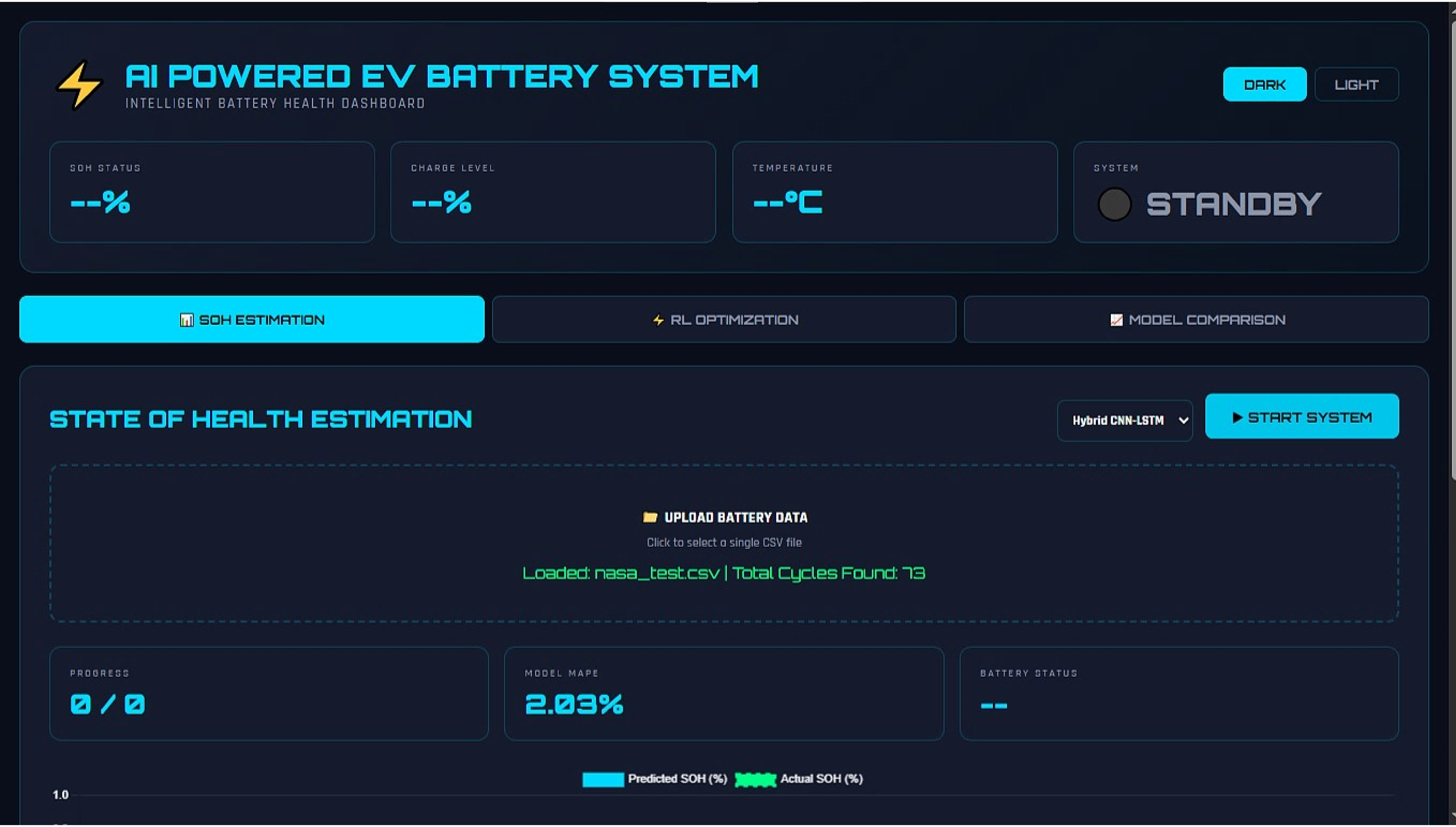

Open
FYGP_Final › estimation › dataset
Search dataset
Organize
New folder
Name
Date modified
Type
Quick access
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Local Disk (C:)
Network
nasa_test
01-05-2026 16:08
Microsoft Excel
nasa_train
01-05-2026 16:08
Microsoft Excel
nasa_val
01-05-2026 16:09
Microsoft Excel
File name:
Microsoft Excel Comma Separa
Open
Cancel


Open
FYGP_Final › estimation › dataset
Search dataset
Organize
New folder
Name
Date modified
Type
Quick access
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Local Disk (C:)
Network
nasa_test 01-05-2026 16:08 Microsoft Excel
nasa_train 01-05-2026 16:08 Microsoft Excel
nasa_val 01-05-2026 16:09 Microsoft Excel
File name: nasa_test
Microsoft Excel Comma Separa
Open
Cancel

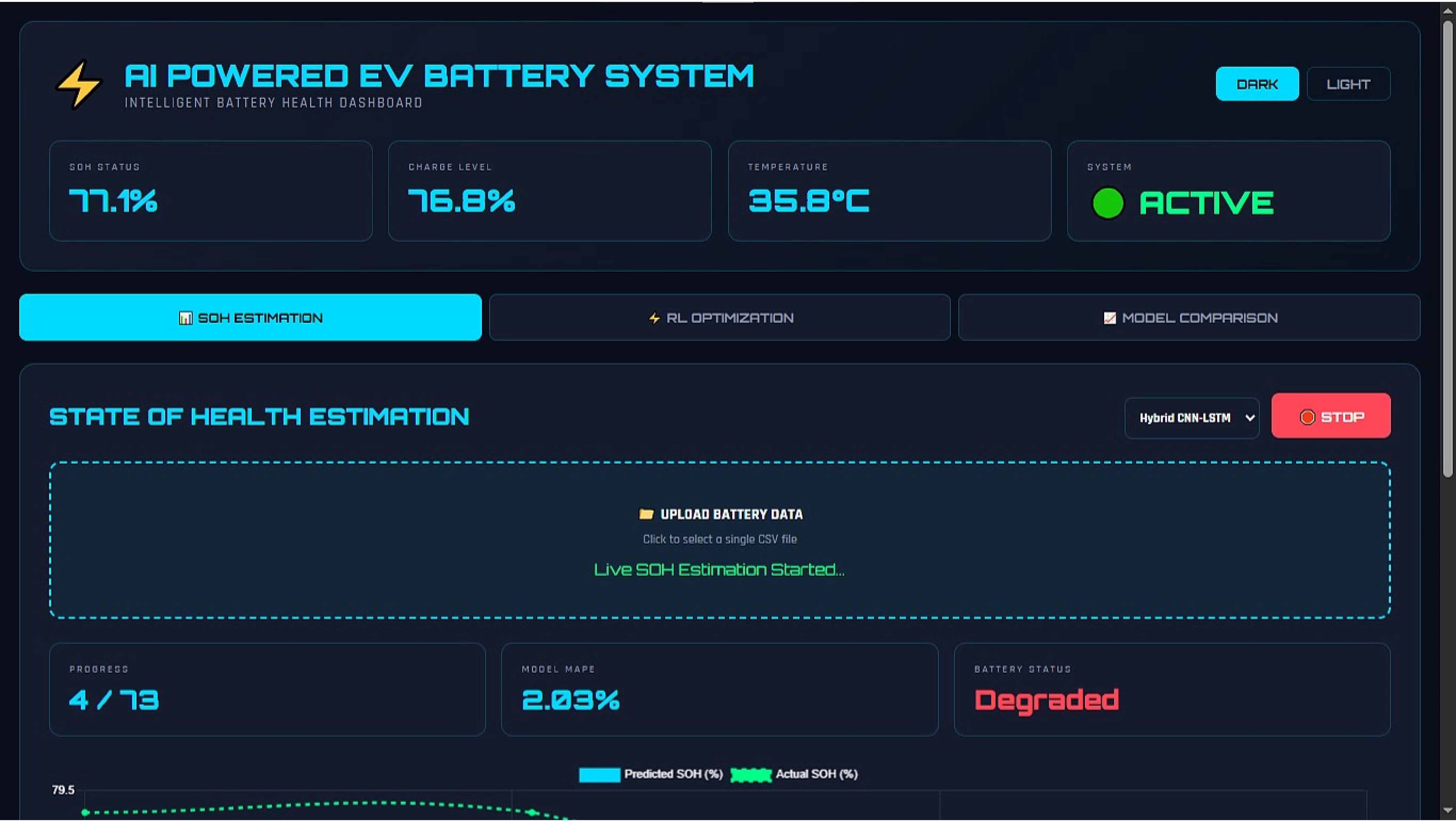

AI POWERED EV BATTERY SYSTEM
INTELLIGENT BATTERY HEALTH DASHBOARD
DARK
LIGHT
SOH STATUS
--%
CHARGE LEVEL
--%
TEMPERATURE
--°C
SYSTEM
STANDBY
SOH ESTIMATION
RL OPTIMIZATION
MODEL COMPARISON
STATE OF HEALTH ESTIMATION
Hybrid CNN-LSTM
START SYSTEM
UPLOAD BATTERY DATA
Click to select a single CSV file
Loaded: nasa_test.csv | Total Cycles Found: 73
PROGRESS
0 / 0
MODEL MAPE
2.03%
BATTERY STATUS
--
Predicted SOH (%)
Actual SOH (%)
1.0


AI POWERED EV BATTERY SYSTEM
INTELLIGENT BATTERY HEALTH DASHBOARD
DARK
LIGHT
SOH STATUS
--%
CHARGE LEVEL
--%
TEMPERATURE
--°C
SYSTEM
STANDBY
SOH ESTIMATION
RL OPTIMIZATION
MODEL COMPARISON
STATE OF HEALTH ESTIMATION
Hybrid CNN-LSTM
START SYSTEM
UPLOAD BATTERY DATA
Click to select a single CSV file
Loaded: nasa_test.csv | Total Cycles Found: 73
PROGRESS
0 / 0
MODEL MAPE
2.03%
BATTERY STATUS
--
Predicted SOH (%)
Actual SOH (%)
1.0


AI POWERED EV BATTERY SYSTEM
INTELLIGENT BATTERY HEALTH DASHBOARD
DARK
LIGHT
SOH STATUS
--%
CHARGE LEVEL
--%
TEMPERATURE
--°C
SYSTEM
STANDBY
SOH ESTIMATION
RL OPTIMIZATION
MODEL COMPARISON
STATE OF HEALTH ESTIMATION
Hybrid CNN-LSTM
START SYSTEM
UPLOAD BATTERY DATA
Click to select a single CSV file
Loaded: nasa_test.csv | Total Cycles Found: 73
PROGRESS
0 / 0
MODEL MAPE
2.03%
BATTERY STATUS
--
Predicted SOH (%)
Actual SOH (%)
1.0


AI POWERED EV BATTERY SYSTEM
INTELLIGENT BATTERY HEALTH DASHBOARD
DARK
LIGHT
SOH STATUS
--%
CHARGE LEVEL
--%
TEMPERATURE
--°C
SYSTEM
STANDBY
SOH ESTIMATION
RL OPTIMIZATION
MODEL COMPARISON
STATE OF HEALTH ESTIMATION
Hybrid CNN-LSTM
START SYSTEM
UPLOAD BATTERY DATA
Click to select a single CSV file
Loaded: nasa_test.csv | Total Cycles Found: 73
PROGRESS
0 / 0
MODEL MAPE
2.03%
BATTERY STATUS
--
Predicted SOH (%)
Actual SOH (%)
1.0


AI POWERED EV BATTERY SYSTEM
INTELLIGENT BATTERY HEALTH DASHBOARD
DARK
LIGHT
SOH STATUS
--%
CHARGE LEVEL
--%
TEMPERATURE
--°C
SYSTEM
ACTIVE
SOH ESTIMATION
RL OPTIMIZATION
MODEL COMPARISON
STATE OF HEALTH ESTIMATION
Hybrid CNN-LSTM
STOP
UPLOAD BATTERY DATA
Click to select a single CSV file
Live SOH Estimation Started...
PROGRESS
0 / 0
MODEL MAPE
2.03%
BATTERY STATUS
--
Predicted SOH (%)
Actual SOH (%)
1.0


AI POWERED EV BATTERY SYSTEM
INTELLIGENT BATTERY HEALTH DASHBOARD
DARK
LIGHT
SOH STATUS
77.1%
CHARGE LEVEL
76.8%
TEMPERATURE
35.8°C
SYSTEM
ACTIVE
SOH ESTIMATION
RL OPTIMIZATION
MODEL COMPARISON
STATE OF HEALTH ESTIMATION
Hybrid CNN-LSTM
STOP
UPLOAD BATTERY DATA
Click to select a single CSV file
Live SOH Estimation Started...
PROGRESS
4 / 73
MODEL MAPE
2.03%
BATTERY STATUS
Degraded
Predicted SOH (%)
Actual SOH (%)
79.5

PROGRESS

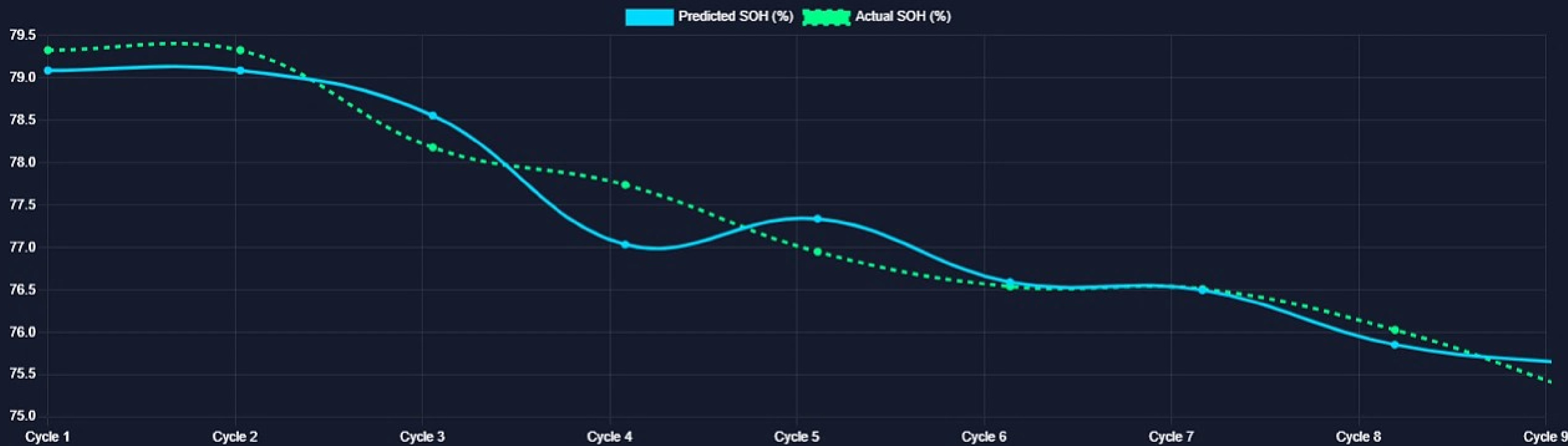
9 / 73

MODEL MAPE

2.03%

BATTERY STATUS

Degraded



Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

Team: Md. Shiban Alam, Syed Sarafeena Ali, Purbasha Roy, Oindrila Sain

Mentor: Mrs. Amrita Bhattacharya | **Institution:** STCET - MAKAUT

PROGRESS

18 / 73

MODEL MAPE

2.03%

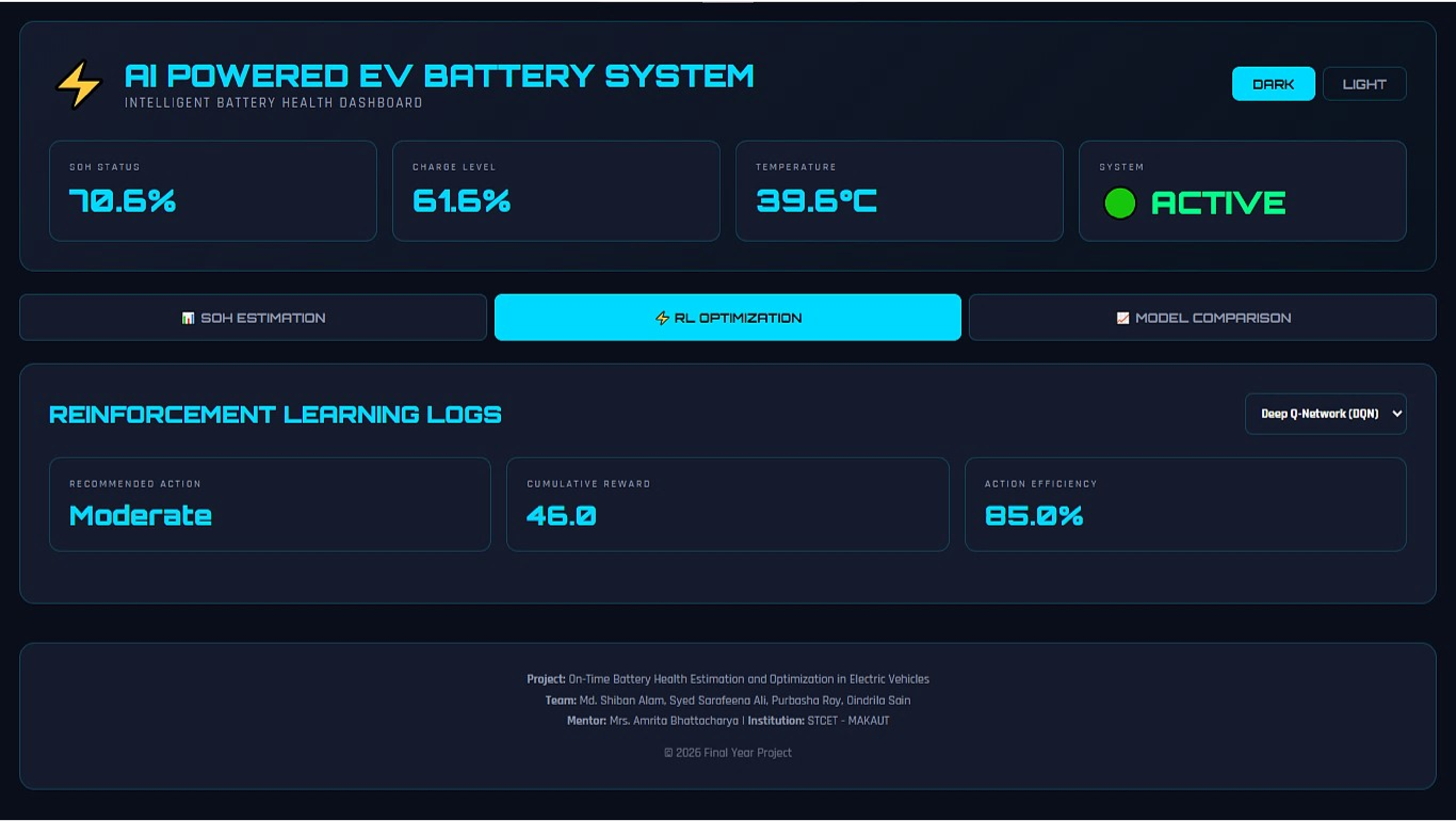
BATTERY STATUS

Degraded

Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

Team: Md. Shiban Alam, Syed Sarafeena Ali, Purbasha Roy, Dindrila Sain

Mentor: Mrs. Amrita Bhattacharya | **Institution:** STCET - MAKAUT


AI POWERED EV BATTERY SYSTEM
INTELLIGENT BATTERY HEALTH DASHBOARD
DARK
LIGHT
SOH STATUS
70.6%
CHARGE LEVEL
61.6%
TEMPERATURE
39.6°C
SYSTEM
ACTIVE
SOH ESTIMATION
RL OPTIMIZATION
MODEL COMPARISON
REINFORCEMENT LEARNING LOGS
Deep Q-Network (DQN)
RECOMMENDED ACTION
Moderate
CUMULATIVE REWARD
46.0
ACTION EFFICIENCY
85.0%
Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles
Team: Md. Shiban Alam, Syed Sarafeena Ali, Purbasha Roy, Oindrila Sain
Mentor: Mrs. Amrita Bhattacharya | Institution: STCET - MAKAUT
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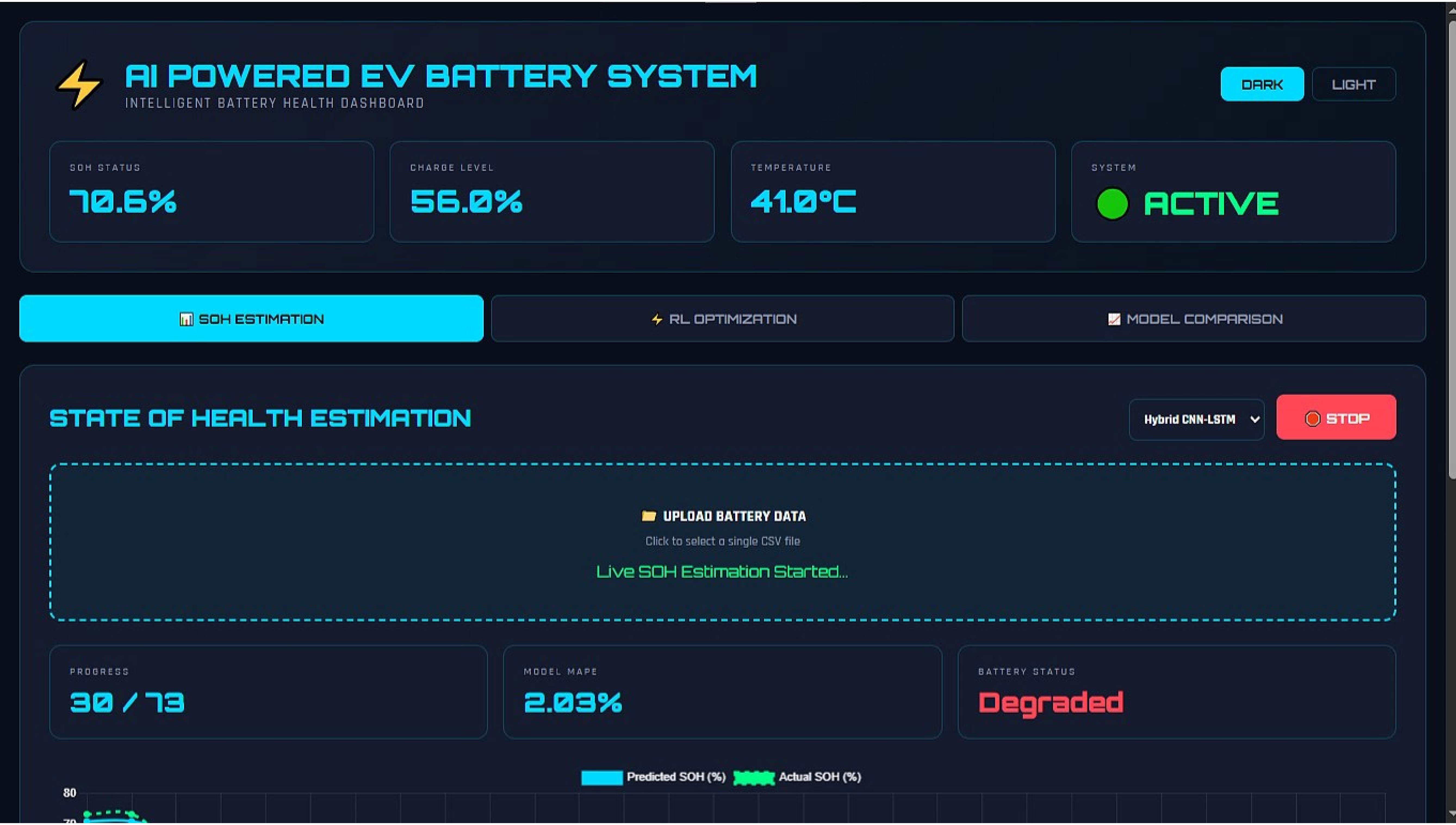
AI POWERED EV BATTERY SYSTEM
INTELLIGENT BATTERY HEALTH DASHBOARD
DARK
LIGHT
SOH STATUS
70.6%
CHARGE LEVEL
56.0%
TEMPERATURE
41.0°C
SYSTEM
ACTIVE
SOH ESTIMATION
RL OPTIMIZATION
MODEL COMPARISON
STATE OF HEALTH ESTIMATION
Hybrid CNN-LSTM
STOP
UPLOAD BATTERY DATA
Click to select a single CSV file
Live SOH Estimation Started...
PROGRESS
30 / 73
MODEL MAPE
2.03%
BATTERY STATUS
Degraded
Predicted SOH (%)
Actual SOH (%)
80

PROGRESS

33 / 73

MODEL MAPE

2.03%

BATTERY STATUS

Degraded

Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

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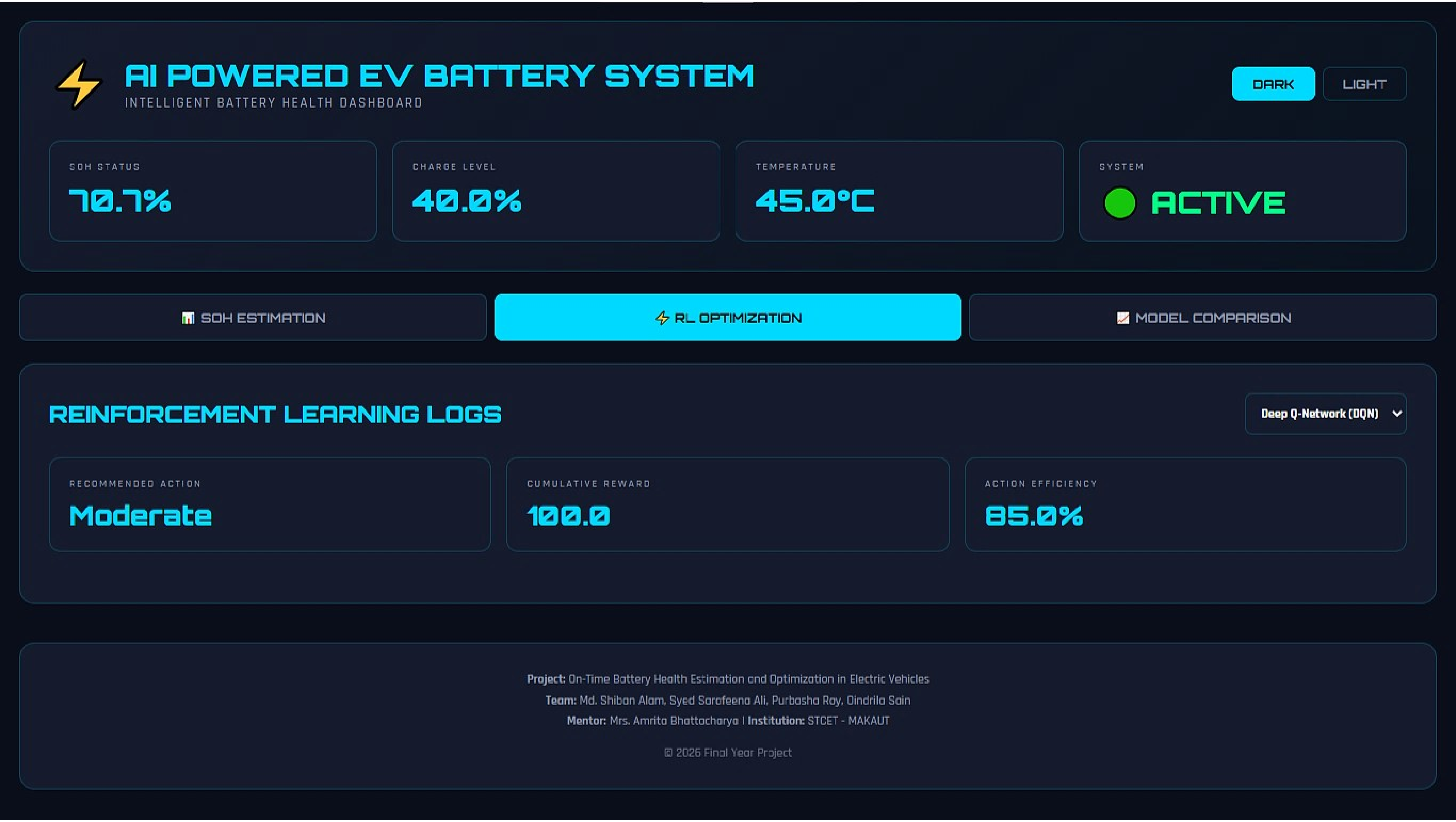
AI POWERED EV BATTERY SYSTEM
INTELLIGENT BATTERY HEALTH DASHBOARD
DARK
LIGHT
SOH STATUS
70.7%
CHARGE LEVEL
40.0%
TEMPERATURE
45.0°C
SYSTEM
ACTIVE
SOH ESTIMATION
RL OPTIMIZATION
MODEL COMPARISON
REINFORCEMENT LEARNING LOGS
Deep Q-Network (DQN)
RECOMMENDED ACTION
Moderate
CUMULATIVE REWARD
100.0
ACTION EFFICIENCY
85.0%
Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles
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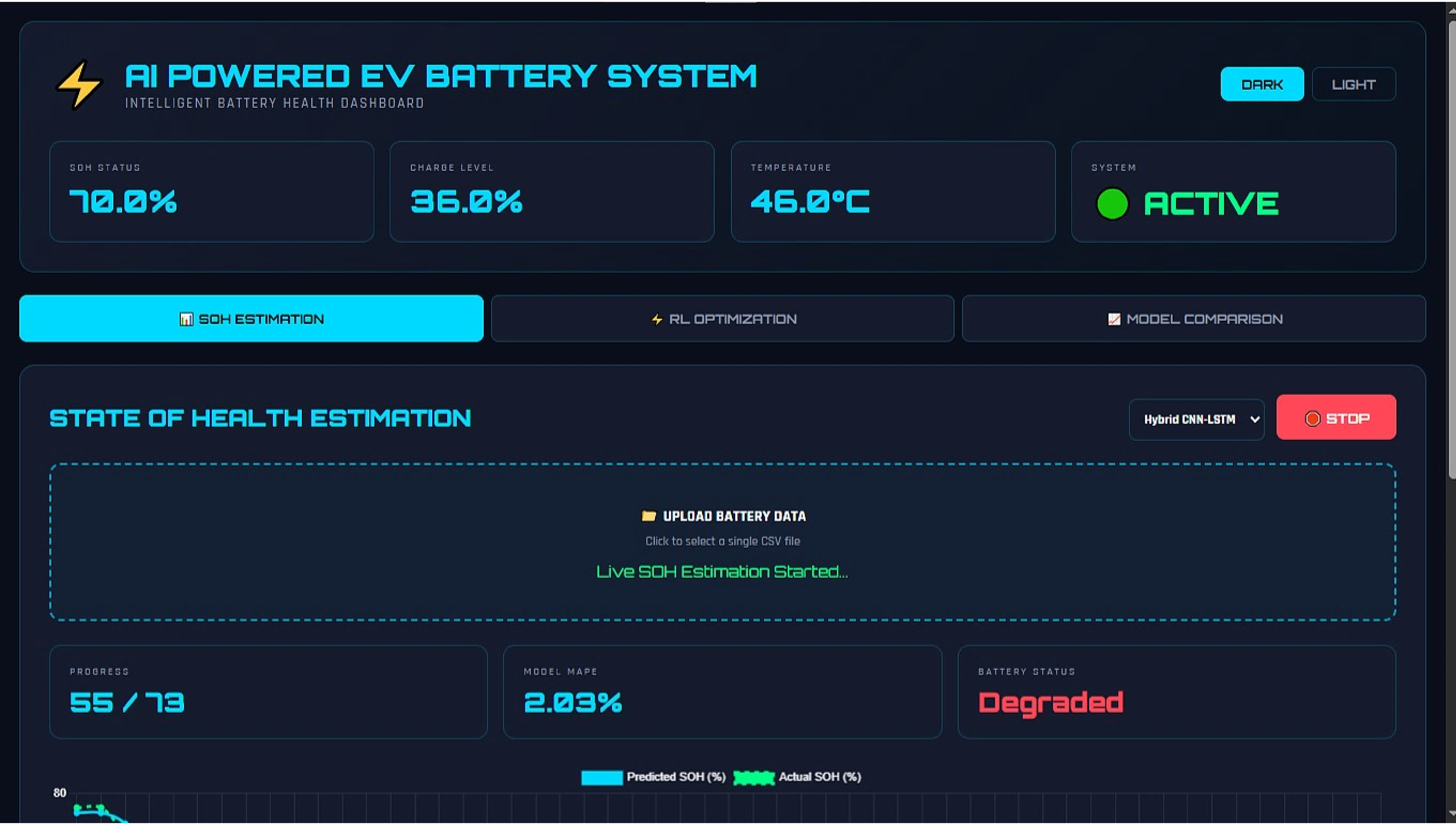
AI POWERED EV BATTERY SYSTEM
INTELLIGENT BATTERY HEALTH DASHBOARD
DARK
LIGHT
SOH STATUS
70.0%
CHARGE LEVEL
36.0%
TEMPERATURE
46.0°C
SYSTEM
ACTIVE
SOH ESTIMATION
RL OPTIMIZATION
MODEL COMPARISON
STATE OF HEALTH ESTIMATION
Hybrid CNN-LSTM
STOP
UPLOAD BATTERY DATA
Click to select a single CSV file
Live SOH Estimation Started...
PROGRESS
55 / 73
MODEL MAPE
2.03%
BATTERY STATUS
Degraded
Predicted SOH (%)
Actual SOH (%)
80

PROGRESS

62 / 73

MODEL MAPE

2.03%

BATTERY STATUS

Degraded

Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

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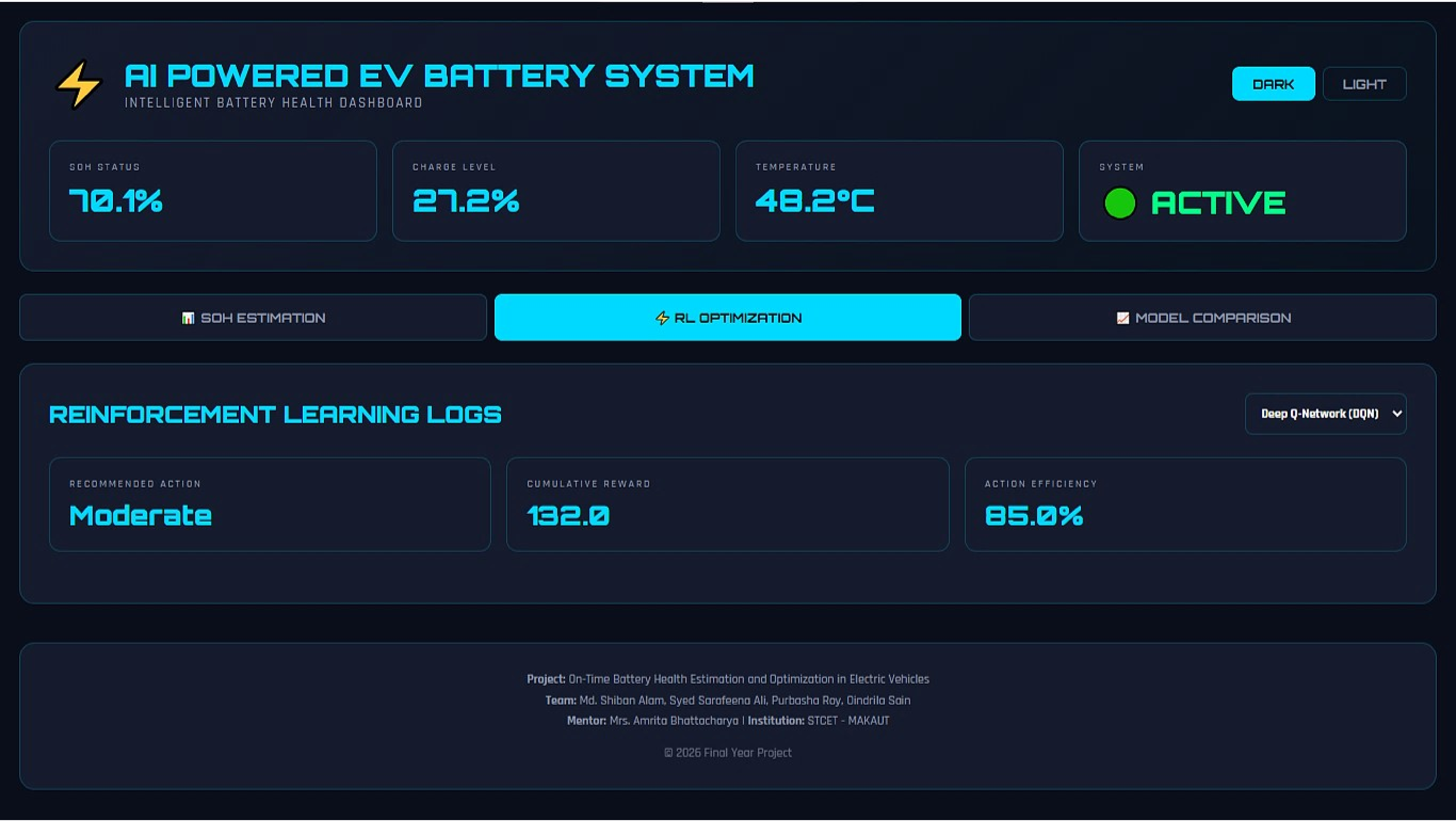
AI POWERED EV BATTERY SYSTEM
INTELLIGENT BATTERY HEALTH DASHBOARD
DARK
LIGHT
SOH STATUS
70.1%
CHARGE LEVEL
27.2%
TEMPERATURE
48.2°C
SYSTEM
ACTIVE
SOH ESTIMATION
RL OPTIMIZATION
MODEL COMPARISON
REINFORCEMENT LEARNING LOGS
Deep Q-Network (DQN)
RECOMMENDED ACTION
Moderate
CUMULATIVE REWARD
132.0
ACTION EFFICIENCY
85.0%
Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles
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AI POWERED EV BATTERY SYSTEM
INTELLIGENT BATTERY HEALTH DASHBOARD
DARK
LIGHT
SOH STATUS
70.9%
CHARGE LEVEL
21.6%
TEMPERATURE
49.6°C
SYSTEM
PROCESSED
SOH ESTIMATION
RL OPTIMIZATION
MODEL COMPARISON
STATE OF HEALTH ESTIMATION
Hybrid CNN-LSTM
START SYSTEM
UPLOAD BATTERY DATA
Click to select a single CSV file
Route Finished: All Telemetry Processed.
PROGRESS
73 / 73
MODEL MAPE
2.03%
BATTERY STATUS
Degraded
Predicted SOH (%)
Actual SOH (%)
80

PROGRESS

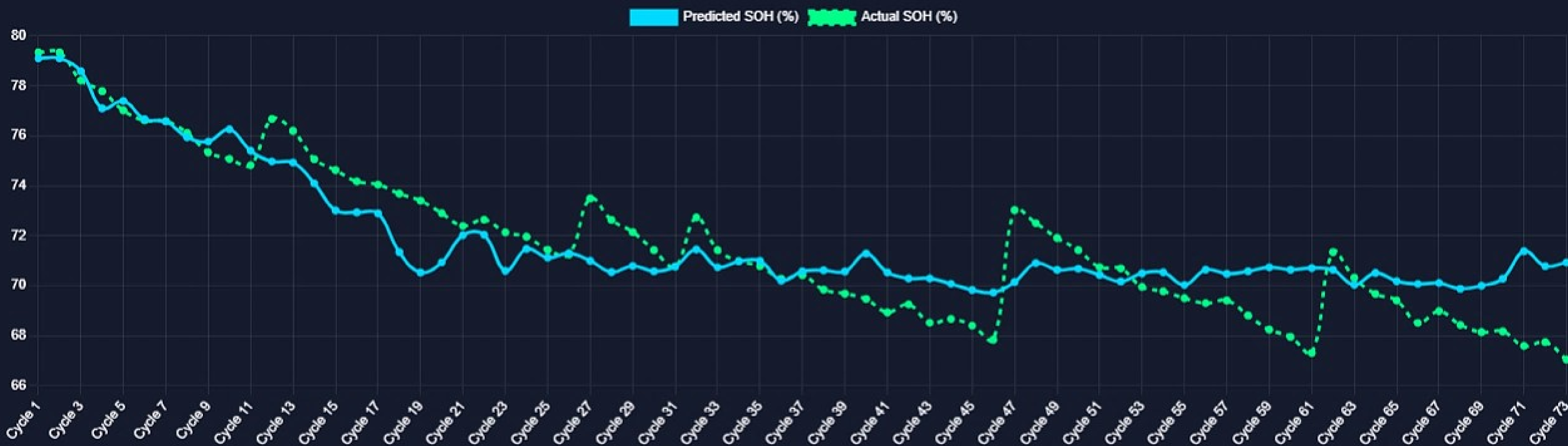
73 / 73

MODEL MAPE

2.03%

BATTERY STATUS

Degraded



Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

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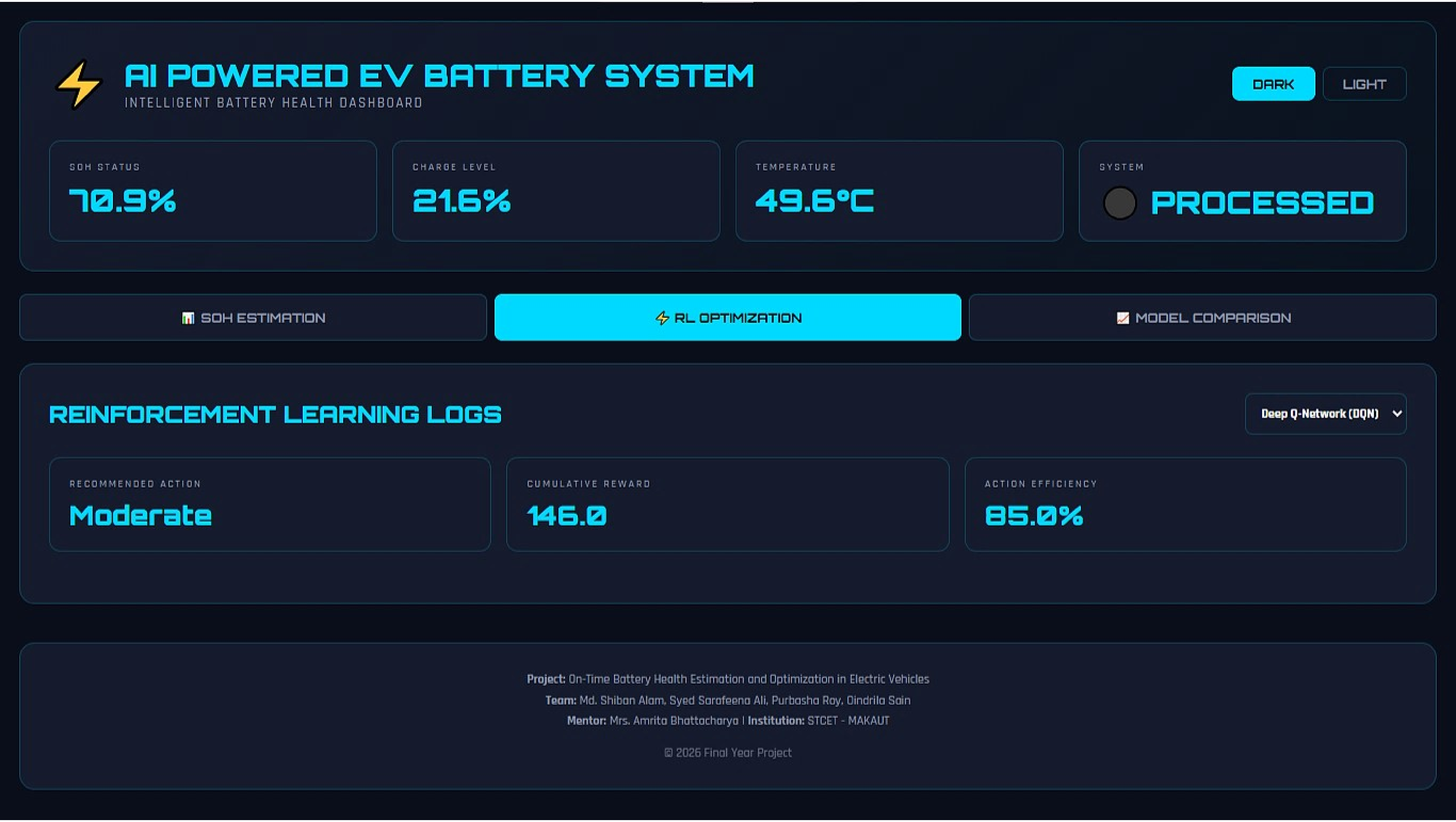
AI POWERED EV BATTERY SYSTEM
INTELLIGENT BATTERY HEALTH DASHBOARD
DARK
LIGHT
SOH STATUS
70.9%
CHARGE LEVEL
21.6%
TEMPERATURE
49.6°C
SYSTEM
PROCESSED
SOH ESTIMATION
RL OPTIMIZATION
MODEL COMPARISON
REINFORCEMENT LEARNING LOGS
Deep Q-Network (DQN)
RECOMMENDED ACTION
Moderate
CUMULATIVE REWARD
146.0
ACTION EFFICIENCY
85.0%
Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles
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